

The Role of a web server is to store files, process requests, send responses, and secure data. The web server would host the components, including HTML pages, CSS sheets, JavaScript files, and images. On my server, it would mainly be HTML, CSS, and images. Then eventually the code written would be sent and read through HTTP or HTTPS requests from a browser. It would locate the original files, which would be on GitHub for me, and send them back to the client. It would secure data using SSL or TLS certificates for safety.

Files are sent to the browser by typing the URL, sending a request, server response, and it would render the page at last. You would enter the right website address into your browser; the browser would check the Domain Name System to find the server's IP address. The browser would send HTTP or HTTPS request to the IP address for specific files and the web server would find the files and send the HTTP response through internet. Lastly, the browser would read the HTML, apply the CSS, run the JavaScript, and display the visual webpage on your screen.

Users would type the web address into the browser address bar. URLs must have https, which helps the browser know how to connect, and a hostname, which would be www.something.com (space added to avoid website links). Once everything is done, the browser would connect and request page data, which would display the website on your screen. All computers and servers on the internet have their own number, which is called an IP address. Domain names are for humans so that we don't need to memorize numbers.

The main differences between HTTP and HTTPS are data security, port numbers, and certificates. HTTP passwords or details can be checked by everyone on the network, while HTTPS scrambles the data for only the correct server and browser. HTTP uses port 80 while HTTPS uses port 443. HTTP doesn't need a special certificate, while HTTPS needs it. My website uses HTTPS because it protects user privacy, secures logins, and it is standard for modern web browsers.

Once all my files were ready I uploaded in GitHub, after I made sure the repository was public then I checked the pages section and made the branch so I could connect through Domain Name.